MILITARY SPECIFICATION

COMPACTOR, HIGH SPEED, SELF-PROPELLED

COMMERCIAL CONSTRUCTION EQUIPMENT (CCE)

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high speed, self-propelled, diesel powered, tamping foot, embankment compactor.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-173	- Varnish, Moisture-and Fungus-Resistant, (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Material.
MIL-R-3075	- Rollers, Motorized, Road, Diesel-or Gasoline-Engine-Driven, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Gun, Hand, High Pressure, Lever-Operated: Cartridge and Bulk Loading (14 Ounce And 21 Ounce Capacity).
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MS35844	- Plug Machine Thread, Magnetic; Drain.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D1557 - Moisture-Density Relations of Soils and Soil-Aggregate Mixtures using 10-lb. (4.54-KG) Rammer and 18-In. (457-MM) Drop.
D2487 - Classification of Soils for Engineering Purposes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

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SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 102 Photochemically Reactive Solvents

(Application for copies should be addressed to the South Coast Air Management District, 9150 E. Flair Drive, El Monte, CA 91731.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The compactor shall be a self-propelled, diesel powered, tamping machine for high speed embankment compaction. The compactor shall be commercial model consisting of articulated steering, a hydraulically controlled strike-off dozer blade, and all wheels shall have tamping foot or segmented impact pad drums with adjustable wheel cleaners. The compactor shall be complete with all components furnished as standard equipment that are customarily offered with a compactor of this type whether stipulated herein or not, together with such attachments as may be necessary to enable the compactor to function reliably and efficiently in a sustained operation. The compactor shall conform to all EPA OSHA legal requirements for US domestic equipment in effect at the date of manufacturer.

3.1.1 Dimensions. The compactor shall conform to the following maximum dimensions:

Length - 29 ft. (without blade).
Width - 12 ft. (without blade).
Height - 12 ft. (exhaust stack removed).

3.1.2 Performance. The compactor shall be capable of compacting at least 1500 compacted cubic yards per hour of soil conforming to type SC (ASTM D2487 Soil Classification Chart) having a plasticity index (PI) of between 10 and 25 inclusive, to at least 95 percent Modified AASHO in compacted lifts of not less than 6 inches. The compactor shall at a minimum, be capable of negotiating a 10 percent grade over material offering a rolling resistance of 150 pounds per ton or greater, for a total effective grade of 17.5 percent. The compactor shall be capable of operating in ambient temperatures up to 125° F without exceeding the engine manufacturer's recommended limits for radiator top tank temperature, engine oil temperatures in the sump and gallery, and hydraulic component manufacturer's fluid temperature limits.

3.2 First article (preproduction model). Unless otherwise specified (see 6.2), the contractor shall furnish one or more compactors for examination and

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testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce compactors that comply with the requirements of this specification. Examinations and tests shall be as specified in Section 4 and unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Production compactors. All compactors furnished under this specification shall be physically and mechanically identical to the first article compactor approved by the Government (see 6.3).

3.4 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.7). Materials, coatings and special treatments utilized shall provide protection against the various forms of material deterioration in the applicable operating and storage environments to which the compactor may be exposed.

3.4.1 Dissimilar Metals. Dissimilar metals as defined in MIL-T-704 shall be insulated from each other to preclude or minimize galvanic corrosion.

3.5 Maintenance.

3.5.1 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement with minimum removal of other major assemblies and installed attachments. Covers or plates which must be removed for component adjustment or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be accessible. Ease of maintenance provisions shall incorporate features insuring operating clearances for handtools and lubricating equipment, and the accessibility for service operations such as adding oil to the engine crankcase and adding oil to gear cases. Each maintenance, assembly, or disassembly operation required during testing in accordance with 4.3 shall be accomplished by no more than two men (except a hoist and operator may be used as necessary) using special tools furnished with the compactor and common tools. After completion of any maintenance operation, the compactor shall operate as specified herein.

3.5.2 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Common hardware and fastening devices such as bolts, nuts, screws, and washers shall be held to a minimum number of types and sizes. All screw threads shall be in accordance with FED-STD-H28.

3.6 Safety and sound level.

3.6.1 Safety. All exposed parts which are subject to high operating temperatures, or which are energized electrically shall be insulated, fully enclosed or guarded. All rotating, reciprocating or moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. All walking surfaces shall be provided with

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an anti-skid surface. Unless otherwise specified (see 6.2), seat belts shall be provided and the belts shall be anchored in accordance with SAE J386. Operator visibility shall be adequate to permit safe operation in forward and reverse travel. All platforms, steps, and hand holds shall be in accordance with SAE J185 and permit unhampered and non-hazardous entrance and exit from the equipment. When specified (see 6.2), a reverse alarm of the audible type shall be provided.

3.6.2 Sound level. The exterior sound level of the compactor shall not exceed 88dB(A) when measured in accordance with SAE J88. If the noise level in the area occupied by the operator is 85 dB(A) or greater when measured in accordance with SAEJ919, a caution plate shall be permanently affixed on the compactor in a conspicuous protected location and shall be clearly visible and legible to all personnel exposed to the excessive noise levels. The caution plate shall read: "CAUTION HEARING PROTECTION REQUIRED".

3.6.3 Rollover protective structure (ROPS). When specified (see 6.2), the compactor shall be equipped with a removable bolted down ROPS. The ROPS shall conform to SAE J1040 and SAE J231. A slinging eye(s) shall be provided to facilitate ROPS removal.

3.7 Frame. The frame shall be of the articulated type. The frame or wheel assembly shall be constructed to allow wheel oscillation.

3.7.1 Steering system. The compactor shall be equipped with a hydraulic powered articulated (center point) type steering system which provides for tracking of front and rear wheels or drums or full width coverage. Compactors, built in two sections with a hingepoint for articulation, shall have steering frame locks conforming to SAE J276. A caution plate shall be posted on each side of the compactor near the articulation point cautioning personnel not to stand in the area of the articulation point when the engine is running and to use the steering locks when servicing the equipment and during transport.

3.8.2 Lighting system. The compactor shall be furnished with front and rear mounted headlamps or floodlamps for illumination of the area in front of and behind the compactor for safe and efficient night operation of the compactor.

3.9 Propulsion system. The compactor shall be equipped with a torque converter and a power shift transmission having at least 3 gear speed ratios. A single forward-reverse level shall be provided to control the direction of the compactor for each of the 3 minimum gear speed ratios. Engine power may also be transmitted through a full hydrostatic drive with infinitely variable speed control in forward and reverse. The compactor engine starting system shall be such that the transmission must be placed in neutral position before the engine can be started.

3.10 Brake system. The compactor shall be provided with an air or air-over-hydraulic or a full hydraulic service brake system. Service brakes shall be capable of bringing the compactor to a complete stop within 25 feet from

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a speed of 10 mph on a dry level surface. In addition, a manual or automatic emergency parking brake shall be provided. The parking brake shall be capable of holding, but not necessarily stopping, the compactor on a 15 percent grade with the engine running, for a period of 15 minutes without the assistance of the service brakes, transmission lock, wheel chocks, or any other means.

3.10.1 Brake fluid. When hydraulic brakes are utilized, the brake system shall be filled with silicone brake fluid, automative type, all weather operational and preservation conforming to MIL-S-46176.

3.11 Strike-off (leveling) dozer blade. The leveling blade shall be hydraulically operated and shall have a replaceable cutting edge. The blade width shall be more than the compactor width, but shall not exceed the compactor width by more than 6 inches on each side. The blade height shall be not less than 30 inches. Hydraulic controls shall include lift, lower, float and hold positions.

3.12 Wheels and cleaners. Each compactor wheel or drum shall be provided with tamping feet or segmented impact pads. The tamping foot tips or segmented impact pad caps shall be of the replaceable type, either bolt-on, pin-on or weld on. Cleaners shall be provided fore and aft, on all drums or wheels. The cleaners shall be adjustable and replaceable and shall prevent material build-up regardless of direction of travel.

3.13 Control station. The control station shall be furnished with an operator's seat with backrest. The seat shall be covered with a durable, waterproof covering. The seat shall be located on the station platform in a position that shall provide the operator, while seated in a normal operating position, with an unobstructed view fore and aft, and to either side of the compactor. All controls and instruments necessary to operate the compactor shall be located at the control station. When seated in a normal operating position, the controls shall be within easy reach of the operator and instruments shall be readily visible. All controls and instruments shall be clearly identified for their intended operation and function with permanently affixed identification plates. The identification plates shall meet the requirements specified in 3.23.

3.13.1 Security protection. When specified (see 6.2), the compactor shall be equipped with locking caps to secure the following: fuel tank, hydraulic oil reservoir, engine oil filler, and radiator, unless protected by engine hood and easy access hood, side doors or panels that can be locked.

3.14 Diesel engine. The compactor shall be powered by industrial diesel engine(s) having horsepower, torque, and speed characteristics to satisfactorily meet all performance requirements specified herein without exceeding the engine(s) manufacturer's intermittent rated horsepower of the engine(s) furnished. The engine(s) governed speed shall be set and sealed in accordance with the engine manufacturer's recommendations. The engine(s) shall be capable of operating continuously at rated load in an ambient temperature of 125° F

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without exceeding the maximum allowable heat temperatures established by the engine manufacturer. The engine shall be furnished with the following accessories:

a. Ammeter or voltmeter.
b. Oil pressure indicator.
c. Oil filter.
d. Air cleaner, dry type.
e. Fuel gauge.
f. High temperature cut off switch.
g. Low oil pressure cut off switch.

3.14.1 Electrical system. The compactor shall be furnished with a 12 volt or 24 volt electrical cranking system including the following components:

a. Batteries.
b. Starter.
c. Alternator.
d. Voltage regulator.
e. Ignition switch.

3.14.2 Engine fuel tank. The fuel tank shall be of sufficient capacity for not less than 8 hours operation at the maximum fuel rating of the engine.

3.14.3 Engine oil sampling valve. An oil sampling valve shall be provided on the engine. The valve shall be manually operated and shall automatically close after release. It shall be made of material resistant to corrosion such that it will not contaminate the sample. The discharge port of the valve shall be covered with a captive chained cap conforming to SAE J514. The valve shall be located in such a way as to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running.

3.15 Electromagnetic interference. Unless otherwise specified (see 6.2), the electromagnetic interference emission and susceptability characteristics of the compactor shall conform to SAE J551, devices category.

3.16 Lifting attachments. Permanently affixed attachments that enable the compactor to be lifted in its normal travel position shall be provided. Each attachment shall meet the requirements specified in MIL-STD-209. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.17.

3.17 Tiedown attachments. Permanently affixed attachments that permit fastening the compactor to the floor or deck of a transportation medium shall be provided. The tiedown attachments shall meet the requirements specified in MIL-STD-209. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.16.

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3.18 Towing attachments. Towing attachment(s) shall be provided. The attachment(s) shall be hook(s) on loop(s) suitable for towing (or recovering) the compactor. Instructions for the use of the attachments shall be included on plates made of material specified in 3.23. The instruction plates shall be permanently affixed to the compactor.

3.19 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating. A lubrication chart in accordance with SAE J753 shall be furnished with each compactor.

3.19.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, Type I, II or III. Fittings shall be located in a protected position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a straight hydraulic coupler and 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.19.2 Pressure-release device. An automatic pressure-release device shall be provided where pressure-lubricating equipment could damage grease seals or other parts.

3.19.3 Filling, draining, and checking provisions. Enclosures such as gearcases and transmission housing that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipstick, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with lubricant and for draining. The drain outlet shall be fitted with a drain plug conforming to MS35844. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to a hand held container.

3.19.4 Grease lubrication. All grease lubrication shall be with greases conforming to MIL-G-10924 or MIL-G-23827, as applicable. The compactor shall be assembled, run-in, tested, and delivered with these greases. The bearings shall be cleaned before lubrication because military greases are not compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.19.5 Oil lubrication. Unless otherwise preserved for packaging, all components shall be filled to the operating level with winter grades of military oil designated for use in the temperature range of 0° F to 32° F. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricants used.

3.19.6 Servicing and adjustments. The contractor shall service and adjust each connector for correct operation and travel including at least the

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following: adjustment of the engine; adjustment of the brake system and electrical system; complete lubrication of the compactor with the types and grades of lubricant recommended.

3.20 Fungus and moisture resistance. Unless otherwise specified (see 6.2), the electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.20.1.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.20.1 Composition. Composition I shall be used unless composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 102, South Coast Air-Quality Management District.

3.21 Shipping data plates. Shipping data plates shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material and shall show the silhouette of the compactor in transport position indicating the center of gravity and the location and capacity of the lifting and the tiedown attachments. The recommended lifting and tiedown arrangement including spreader bar arrangement shall be included on these plates. The data plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.22 Identification marking. The compactor shall be identified in accordance with MIL-STD-130. The markings shall be applied to the compactor on a plate conforming to MIL-P-514, Type I, Style 1, Composition C, of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.23 Control identification, safety and warning plates. A plate shall be furnished for each switch, gage, and control and at all points requiring attention for safe and efficient operation of the compactor. The plates shall be labeled by lettering or other appropriate marking with information such as function, direction each control is to move, neutral position of each control, safety instructions, and other warnings or instructions, as necessary. The

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plates shall be anodized aluminum, sunken-etched or raised-relief. Plates that describe safety measures or that warn against operations detrimental to the compactor shall be distinguished by the words "CAUTION" or "WARNING" in red lettering. The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.24 Treatment and painting. The portions of the compactor normally painted shall be cleaned, treated, and painted in accordance with the manufacturer's standard practice, provided it is at least equivalent to MIL-T-704. Unless otherwise specified (see 6.2), the finishing coat shall be lusterless forest green.

3.25 Stenciling. The gross weight of each compactor shall be stenciled on each side of the compactor. "LIFT" and "TIEDOWN", as applicable, shall be stenciled adjacent to their respective tiedown and lifting attachments. All stenciling shall be in block-type letters not less than 1 inch high, and shall be black in color.

3.26 Tool box. The compactor shall be provided with a closed compartment or tool box to carry special tools and accessories required for operation and operator maintenance of the equipment.

3.27 Special tools. Special tools required for maintenance and repair, (if required) shall be supplied by the manufacturer for each item of equipment upon delivery and shall be packed with the equipment.

3.28 Workmanship.

3.28.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.28.2 Bolted connections. All bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lock-washers shall be provided. All nuts, bolts, and screws shall be tight.

3.28.3 Riveted connections. All rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the hole. Rivet heads shall be of an approved shape and shall be concentric with the rivet. Rivet heads shall make full contact with the intended surface. The rivets shall be so fabricated as to develop a joint strength not less than the design value.

3.28.4 Welding. Surfaces to be welded shall be free from foreign matter which would be injurious to the weld. Welding procedures shall be in accordance with a

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nationally recognized code. Welds shall be of sufficient size and shape to develop the full design strength of the parts connected by the welds. Welds shall transmit imposed stresses without permanent deformation or failure when subjected to proof or service loadings.

3.28.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defects which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article preproduction inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article preproduction inspection.

4.3.1 Examination. The compactor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The compactor shall be tested as specified in 4.5.2.1 through 4.5.2.7. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each compactor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Test. Each compactor shall be completely assembled, properly adjusted, and otherwise serviced for operation. Each compactor shall be operated for not less than 2 hours. The compactor shall be driven in both the forward and reverse mode for a total distance of not less than 5 miles. All controls shall be operated as many times as necessary, but not less than 30 times, to determine

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ease of operation, effectiveness, and responsiveness, and that the mechanisms actuated by the controls operate promptly, fully, and without restriction, malfunction, or excessive vibration. Gearcases, bearings, and other moving parts shall be examined for excessive heating or abnormal operation insofar as is practicable and possible. The entire compactor shall be examined for evidence of failure, slippage, or deformation. Failure to pass any phase of this test shall be cause for rejection.

4.5 Inspection procedures.

4.5.1 Examination. The compactor shall be examined for the following defects:

101. Compactor not assembled as specified.
102. Dimensions not as specified.
103. Material not as specified, or dissimilar metals not in compliance with MIL-T-704.
104. Maintainability not as specified.
105. Interchangeability not as specified.
106. Safety devices missing or not as specified.
107. Frame not as specified.
108. Steering system or locking device not as specified.
109. Lighting system not as specified.
110. Propulsion system not as specified.
111. Brake system not as specified.
112. Dozer blade not as specified.
113. Wheels and cleaners not as specified.
114. Control station not as specified.
115. Engine and accessories not as specified.
116. Lifting and tiedown attachments not as specified.
117. Towing attachments not as specified.
118. Lubrication system not as specified.
119. Data plates missing or not as specified.
120. Treatment and painting not as specified.
121. Stenciling not as specified.
122. Tool box not as specified.
123. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The compactor shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test.

4.5.2.2 Performance. Operate the compactor on a fill, built over a natural soil subgrade having an initial density of at least 90 percent Modified AASHO. The fill material shall correspond to group SC, having a plasticity index of 10-25, of the Soil Classification Chart in ASTM D2487. The compacted fill site

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shall be not less than 30 feet wide, with the actual width of the fill being a minimum of 3 feet wider than the compacted area. The fill may be placed prior to the start of compaction operation, during compaction operations, or the contractor may elect to till the natural soil in lieu of placing fill material. When hauling equipment is run over spreaded uncompacted fill material, the fill site shall be scarified or tilled to a uniformly loose uncompacted condition. Spreading of the fill material shall be done with a grader. Spreading runs shall be only that necessary to reach the estimated uncompacted fill lift thickness. Water distribution equipment may be used as required. The material to be compacted shall be of sufficient thickness, initially, to give a compacted lift of at least 6 inches. The laboratory density of the soil to be compacted shall be determined in accordance with ASTM D1557, method optional. When the contractor feels that a compaction rate of not less than 1500 cubic yards of fill to a density of 95 percent Modified AASHO has been accomplished, he may stop compaction operations. Time used in density testing shall not be counted as compaction time. Density measurements shall be taken at points along the test strip using the pattern shown on figure I at intervals of 200 feet. In addition a minimum of 3 samples shall be taken at intervals of 100 feet and at random locations specified by the Government. Density testing shall be accomplished by a nuclear density meter. The nuclear density tester shall have been calibrated against the soil to be compacted. An achieved fill material density of less than 95 percent Modified AASHO or less than 6 inches in depth shall constitute failure of this test. Sufficient measurements shall be taken to determine the in-place quantity of material compacted. The above procedure shall be repeated for a total of 3 times. The average compacted cubic yards shall be determined.

4.5.2.2.1 Failure Criteria. An average compaction rate of less than the compacted cubic yards per hour specified in 3.1.2 and failure to achieve compacted lift thickness of at least 6 inches for each test shall constitute failure of this test.

4.5.2.3 Gradeability. Operate the compactor on a minimum grade of 10 percent over fill material having a rolling resistance of not less than 150 pounds per ton (total effective grade of 17.5 percent). The offeror may, at his option, operate the compactor up a grade of at least 17.5 percent.

4.5.2.3.1 Failure criteria. Inability to negotiate the grade under either condition specified above shall constitute failure of this test.

4.5.2.4 High temperature operation. The contractor shall provide proof (test data or calculation) that the compactor can be operated in ambient temperatures up to and including 125° F without exceeding the engine manufacturer's recommended limits for radiator top tank temperature, engine oil temperatures in the sump and gallery, and hydraulic component manufacturer's fluid temperature limit.

4.5.2.4.1 Failure criteria. Failure to meet this requirement shall constitute failure and acceptance of the equipment shall not be approved until corrective action has been taken and approved by the Government.

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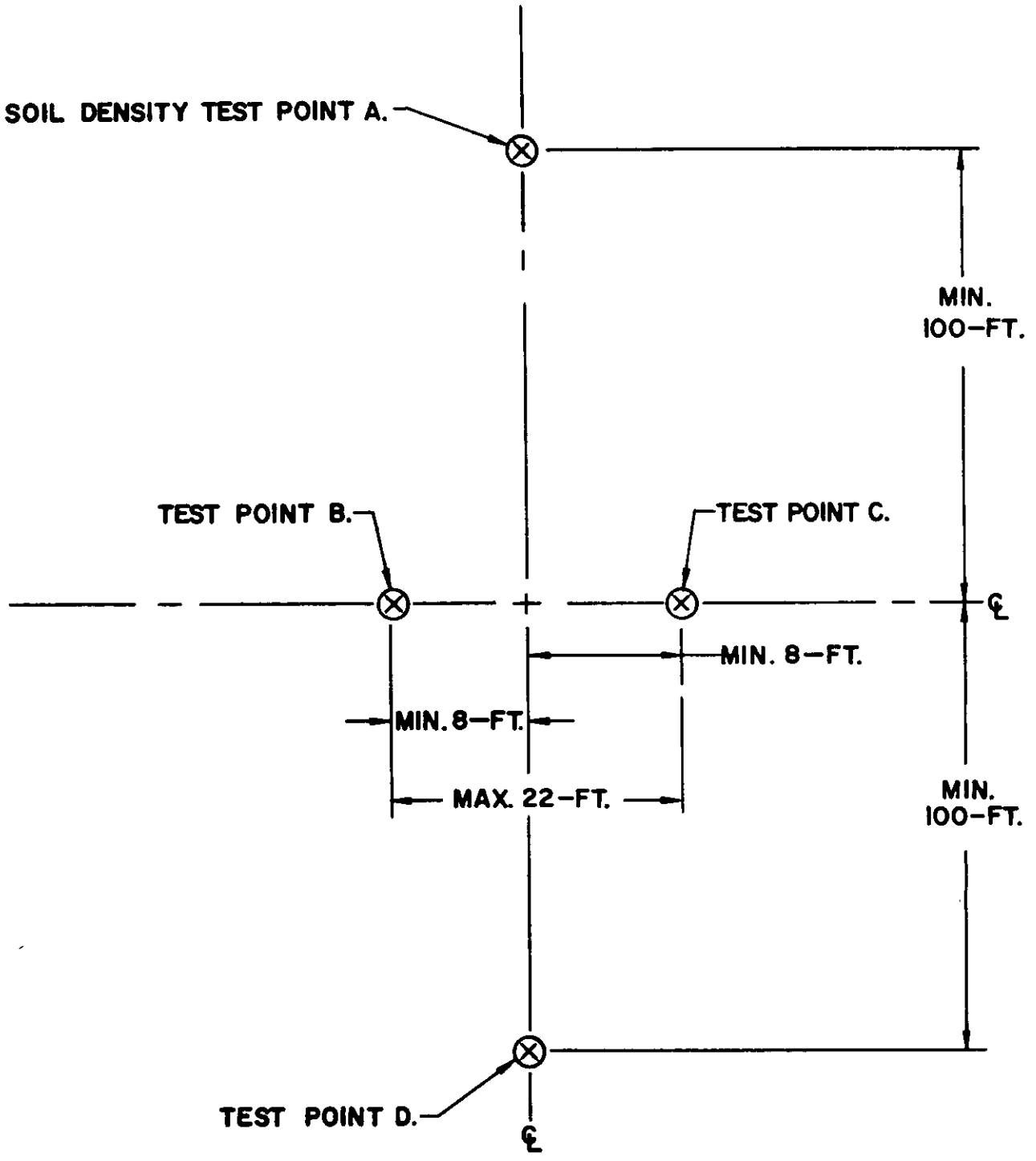


FIGURE 1. Test strip pattern.

X-4080

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4.5.2.5 Electromagnetic interference. Unless otherwise specified, the compactor shall be tested in accordance with SAE J551. A report tabulating the allowable maximum interference and the measured interference shall be furnished with the preproduction model test report.

4.5.2.5.1 Failure criteria. Failure to meet the requirements of SAE J551, shall constitute failure of this test.

4.5.2.6 Sound level measurement.

4.5.2.6.1 Exterior sound level. The contractor shall perform sound level test in accordance with SAE J88.

4.5.2.6.1.1 Failure criteria. Nonconformance to the requirements of 3.6.2 shall constitute failure of this test.

4.5.2.6.2 Operators stations. The compactor shall be tested in accordance with SAE J919.

4.5.2.6.2.1 Failure criteria. Failure to meet the criteria specified in 3.6.2 shall constitute failure of this test.

4.5.2.7 Brake test. The compactor shall be completely stopped five times as specified in 3.10 within a period of 10 minutes, to test brake fadeout. The distance from point of brake application to actual stop shall be measured on the fifth pass to determine conformance of 3.10. The service brake test shall be conducted in both forward and reverse. With the roller stopped on a 15 percent slope, the parking brakes shall be applied and shall be capable of securing the compactor in the stopped position without creeping. The brake test shall be conducted with the compactor heading up and down slope.

4.5.2.7.1 Failure criteria. Failure of the compactor to meet the braking requirements specified in 3.10 shall constitute failure of this test.

4.6 Inspection comparison. The Government may select compactors at any time during the contract production period and subject these compactors to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of these tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison compactor to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison compactor and shall be cause for refusal by the Government to continue

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acceptance of production compactors until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on compactors previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all compactors accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging. The preservation, packing and marking shall be examined in accordance with the Quality Assurance Provisions of MIL-R-3075.

5. PACKAGING

5.1 Preservation, packing and marking. Each complete compactor shall be preserved, packed, and marked, as required for rollers, in accordance with MIL-R-3075. The degree of preservation shall be level A, B, or commercial and the degree of packing shall be level A, B, or commercial, as specified (see 6.2).

6. NOTES

6.1 Intended use. The high speed, segmented compactor covered by this specification will be used for embankment compaction of SC type soils during the construction of roads, airfields, and dams.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. When a preproduction model is not required (see 3.2).
d. Time frame for submission of production model and number of compactors required (see 3.2).
e. When the Government will conduct any or all of the preproduction examination and tests (see 3.2). When the Government will conduct some but not all of the preproduction examination and tests the contracting officer shall specify which examination and test will be conducted by the Government and which examination and test will be conducted by the contractor (see 3.2).
f. When seat belts are not required (see 3.6.1).
g. When a reverse alarm system is required (see 3.6.1).
h. When a rollover protective structure (ROPS) is required (See 3.6.3).
i. When security protection is required (see 3.13.1).
j. When electromagnetic interference requirements are other than as specified (see 3.15).
k. When fungus and moisture resistance is not required (see 3.20).
l. When the finishing coat shall be other than lusterless forest green (see 3.24).
m. Degree of preservation and degree of packing required (see 5.1).

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6.3 Preproduction model. Any changes or deviations of production compactors from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish compactors conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractors maintenance and operation manual to be furnished with each compactor.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of compactors.

6.6 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the compactors in accordance with this specification.

6.7 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodians:
Army - ME

Preparing activity:
Army - ME

Review activity:
Army - AT

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	☐ VENDOR
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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